Name
Date
Course

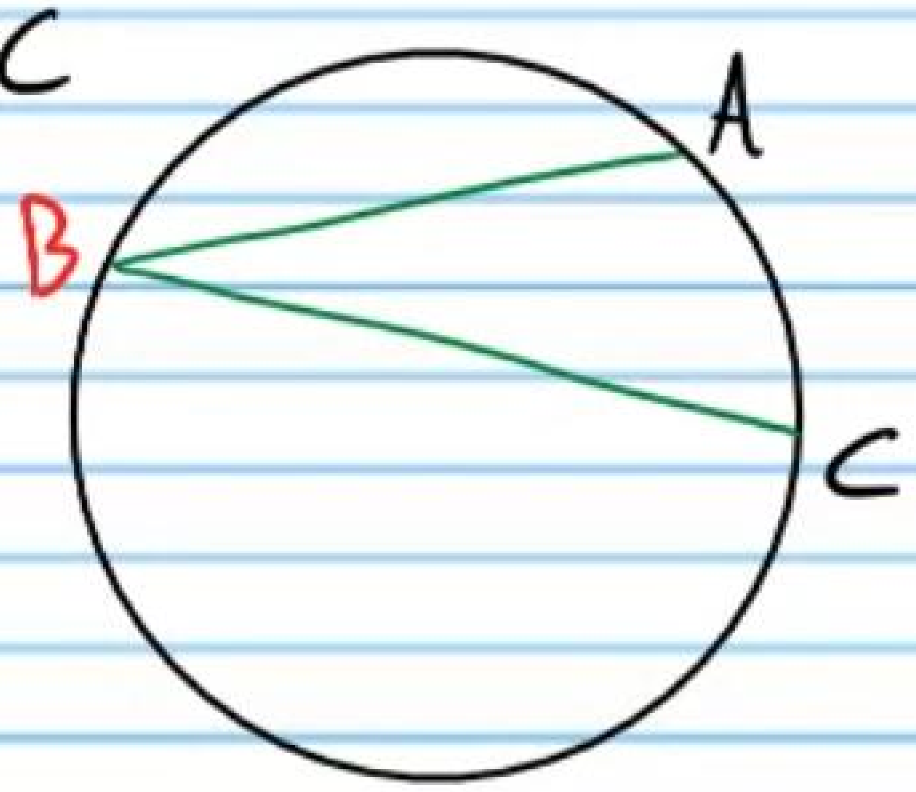
Circles (Part III): Angle and Segment Relationships in Circles

Angle Relationships

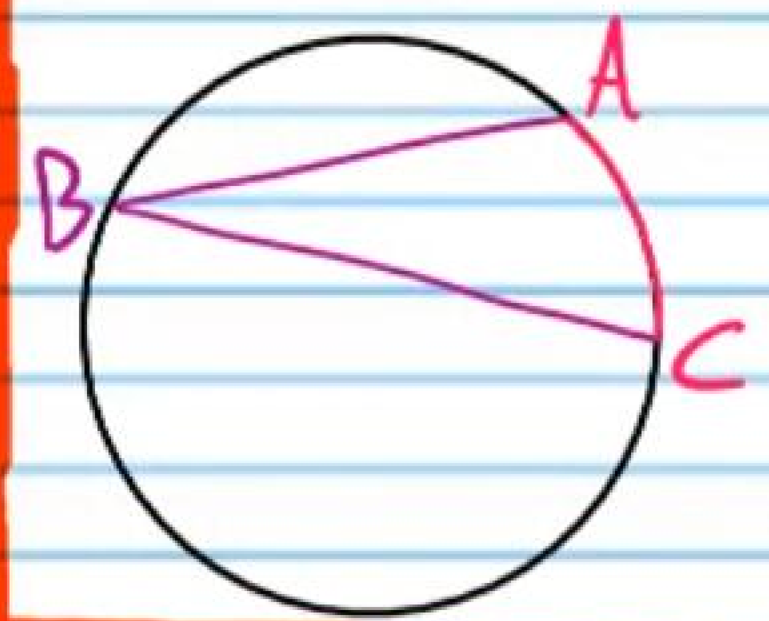
Angles with Vertices on Circles

Inscribed Angle: An angle whose vertex is on a circle and whose sides are chords of the circle.

Example: $\angle ABC$



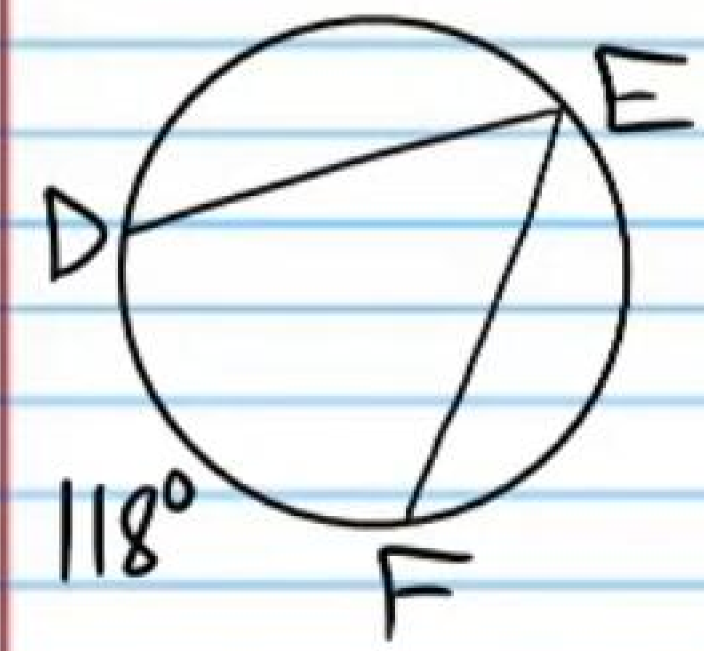
Inscribed Angle Theorem: The measure of an inscribed angle is half the measure of the corresponding arc (also called the intercepted arc).



$$m\angle ABC = \frac{m\widehat{AC}}{2}$$

Find following values.

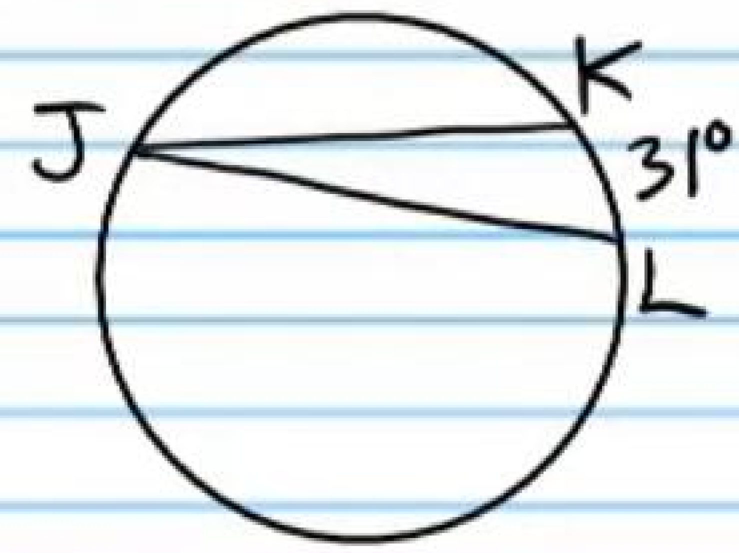
① $m\angle E$



$$m\angle E = \frac{118}{2}$$

$$\boxed{m\angle E = 59^\circ}$$

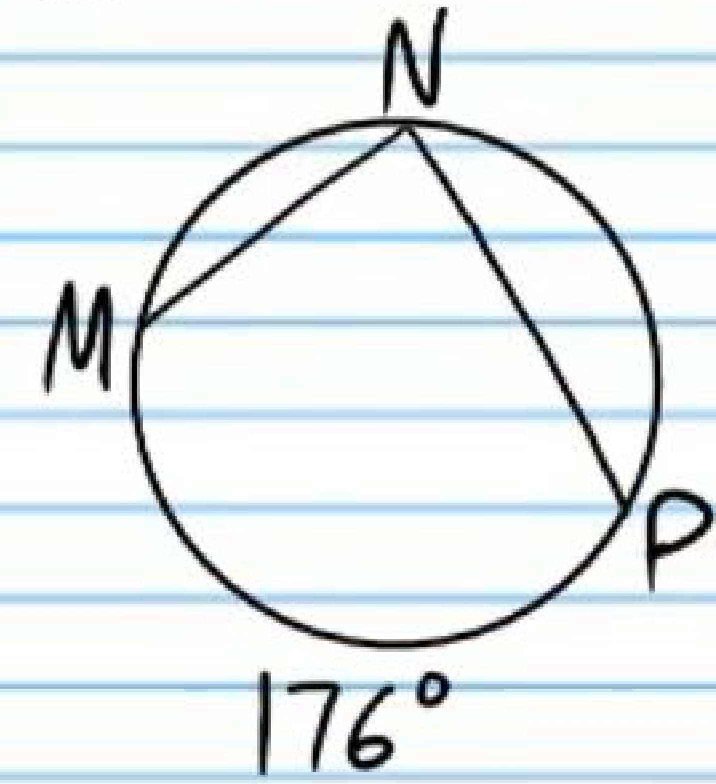
② $m\angle J$



$$m\angle J = \frac{31}{2}$$

$$\boxed{m\angle J = 15.5^\circ}$$

③ $m\angle N$



$$m\angle N = \frac{176}{2}$$

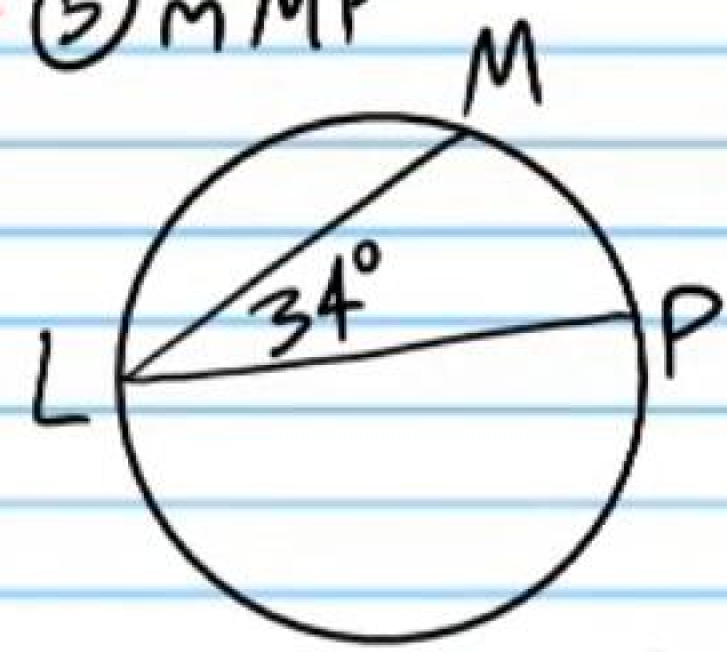
$$\boxed{m\angle N = 88^\circ}$$

④ $m\angle R$

$$m\angle R = \frac{57}{2}$$

$$\boxed{m\angle R = 28.5^\circ}$$

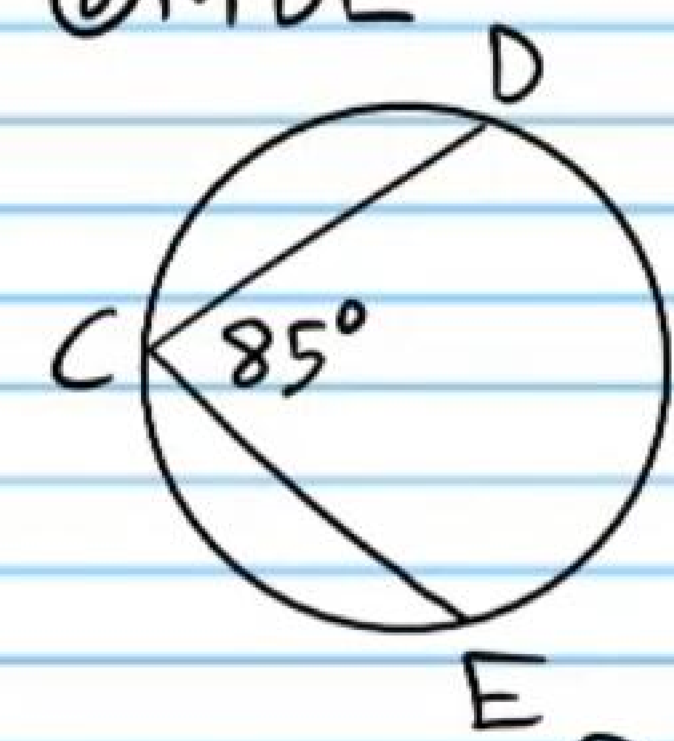
⑤ $m\widehat{MP}$



$$34 = \frac{m\widehat{MP}}{2}$$

$$\boxed{68^\circ = m\widehat{MP}}$$

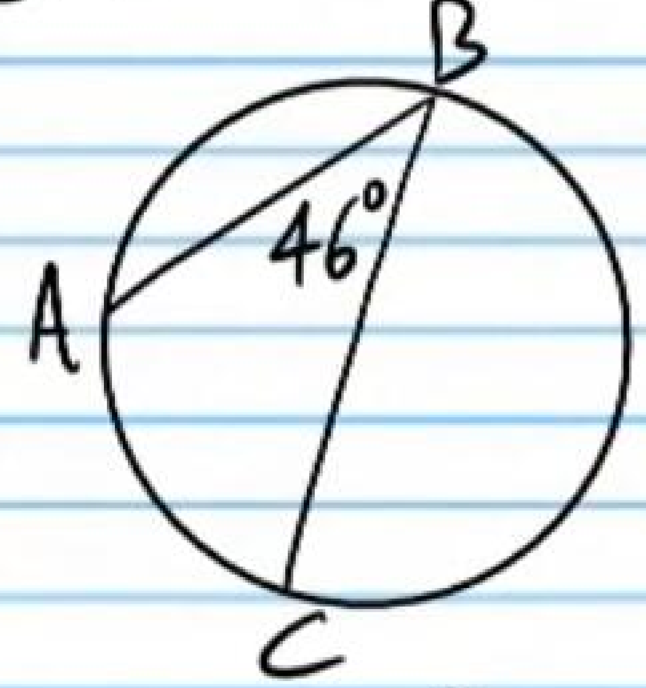
⑥ $m\widehat{DE}$



$$85 = \frac{m\widehat{DE}}{2}$$

$$\boxed{170^\circ = m\widehat{DE}}$$

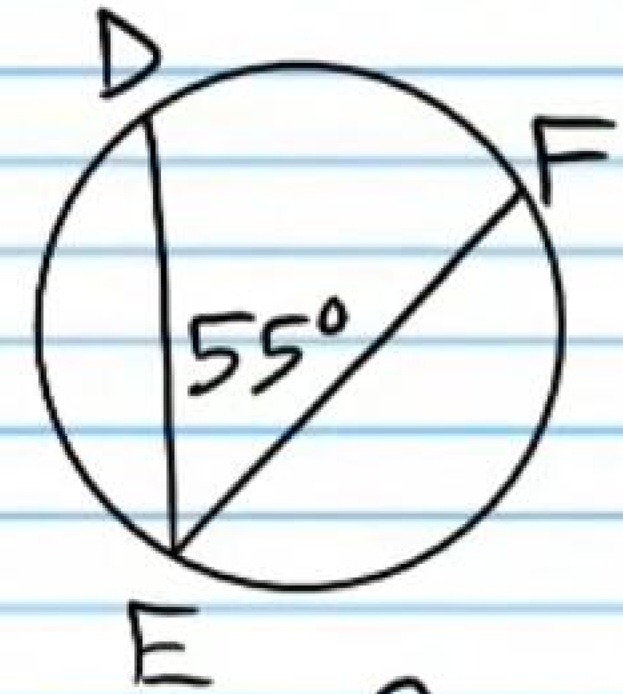
⑦ $m\widehat{AC}$



$46 = \frac{m\widehat{AC}}{2}$

$\boxed{92° = m\widehat{AC}}$

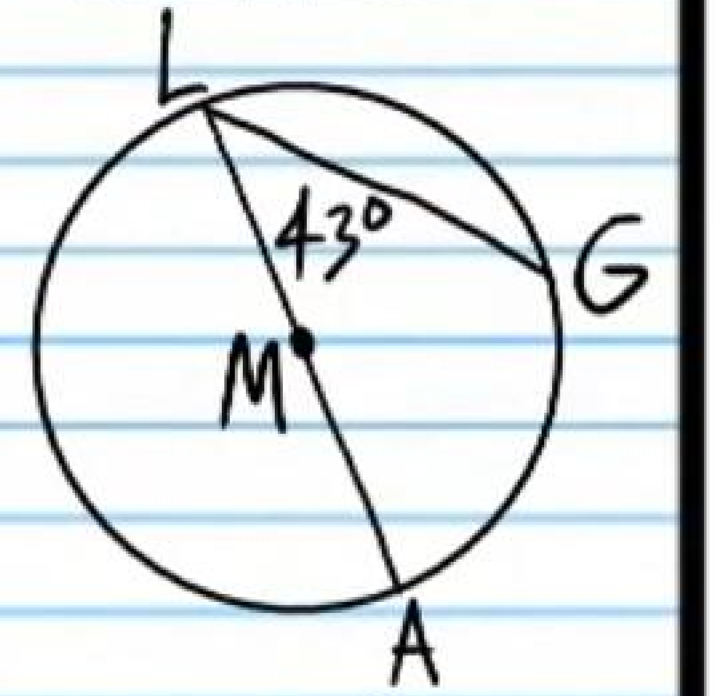
⑧ $m\widehat{DF}$



$55 = \frac{m\widehat{DF}}{2}$

$\boxed{110° = m\widehat{DF}}$

⑨ M is the center. Find $m\widehat{LG}$.

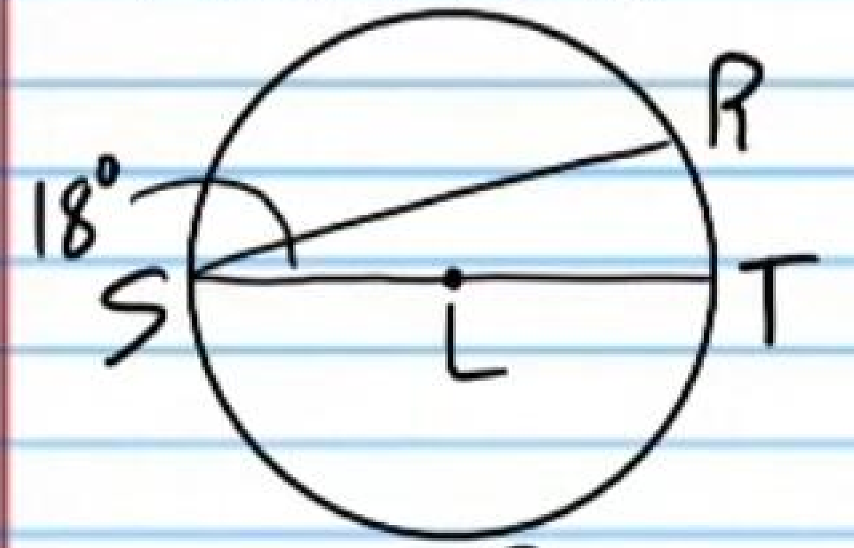


$43 = \frac{m\widehat{AG}}{2}$

$86° = m\widehat{AG}$

$m\widehat{LG} = 180 - 86 = \boxed{94°}$

⑩ L is the center. Find $m\widehat{SR}$.



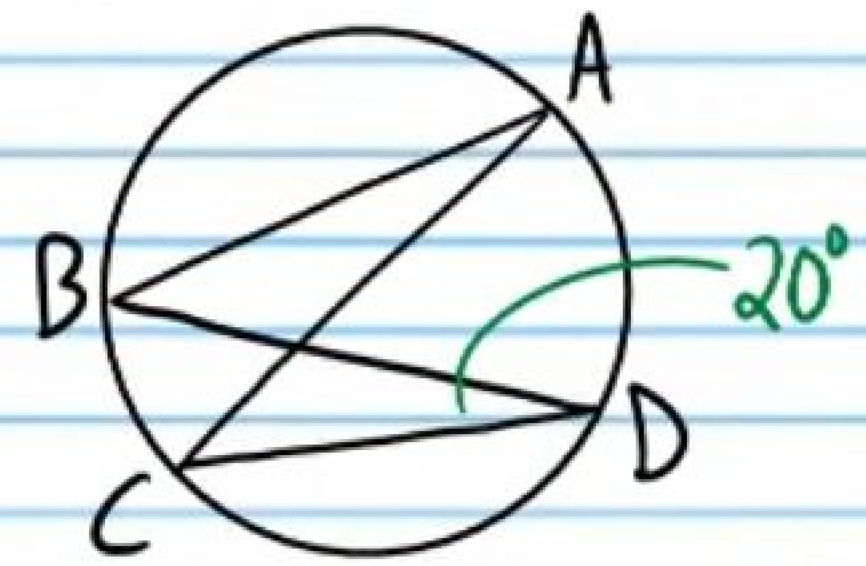
$18 = \frac{m\widehat{RT}}{2}$

$36° = m\widehat{RT}$

$m\widehat{SR} = 180 - 36$

$\boxed{m\widehat{SR} = 144°}$

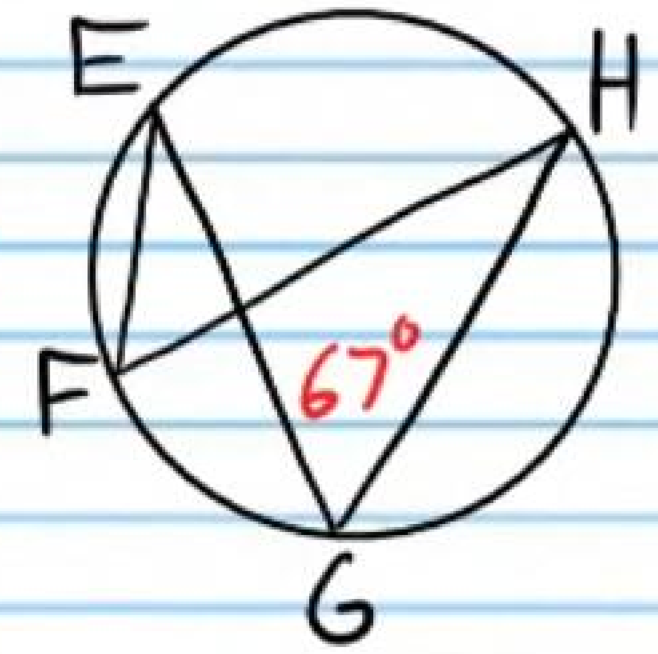
⑪ $m\angle A$



$m\angle A = m\angle D$

$\boxed{m\angle A = 20°}$

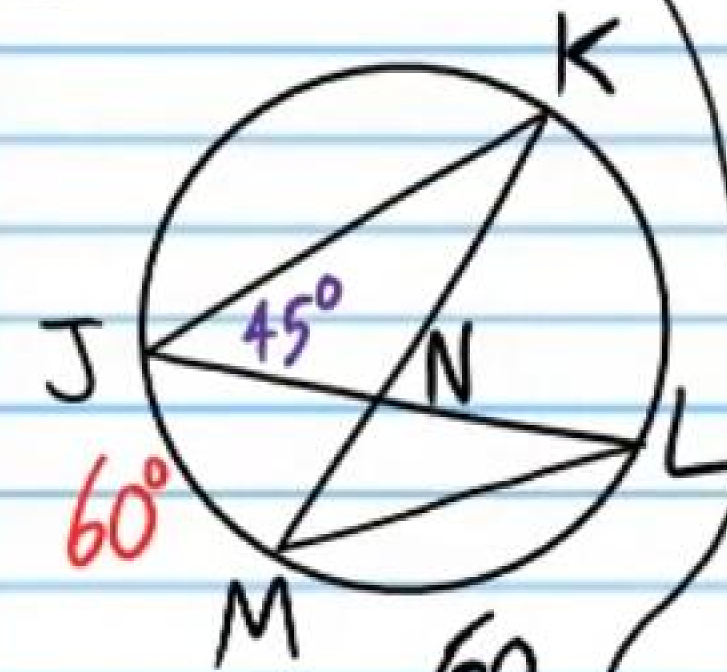
⑫ $m\angle F$



$m\angle F = m\angle G$

$\boxed{m\angle F = 67°}$

⑬ $m\angle JNK$



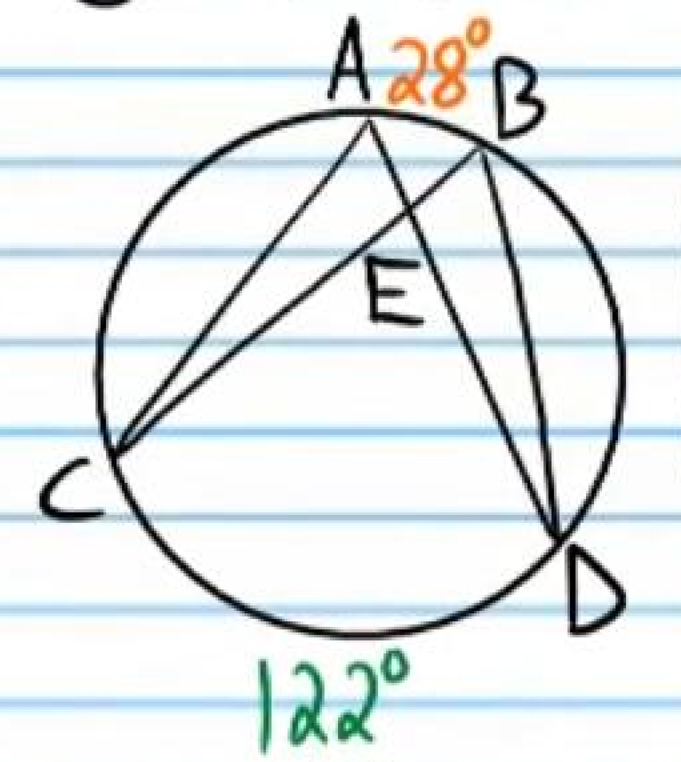
$m\angle K = \frac{60}{2}$

$m\angle K = 30°$

$m\angle JNK = 180 - 30 - 45$

$\boxed{m\angle JNK = 105°}$

⑭ $m\angle DEB$



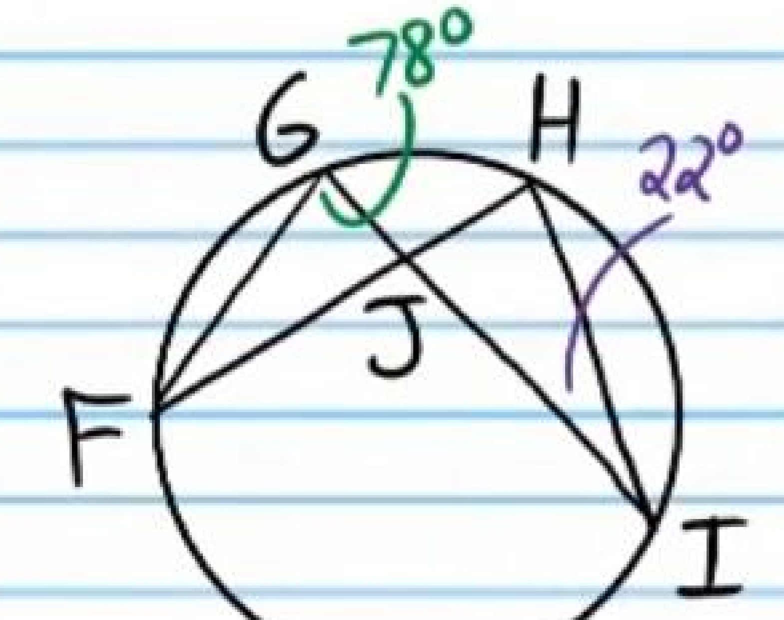
$m\angle B = \frac{122}{2} = 61°$

$m\angle D = \frac{28}{2} = 14°$

$m\angle DEB = 180 - 61 - 14$

$\boxed{m\angle DEB = 105°}$

⑮ $m\angle GJH$



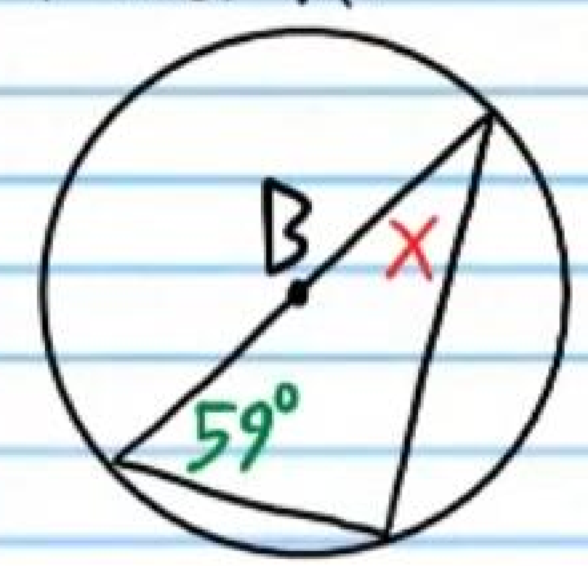
$m\angle GJF = 180 - 78 - 22$

$m\angle GJF = 80°$

$m\angle GJH = 180 - 80$

$\boxed{m\angle GJH = 100°}$

⑯ B is the center. Find X.



$X = 180 - 59 - 90$

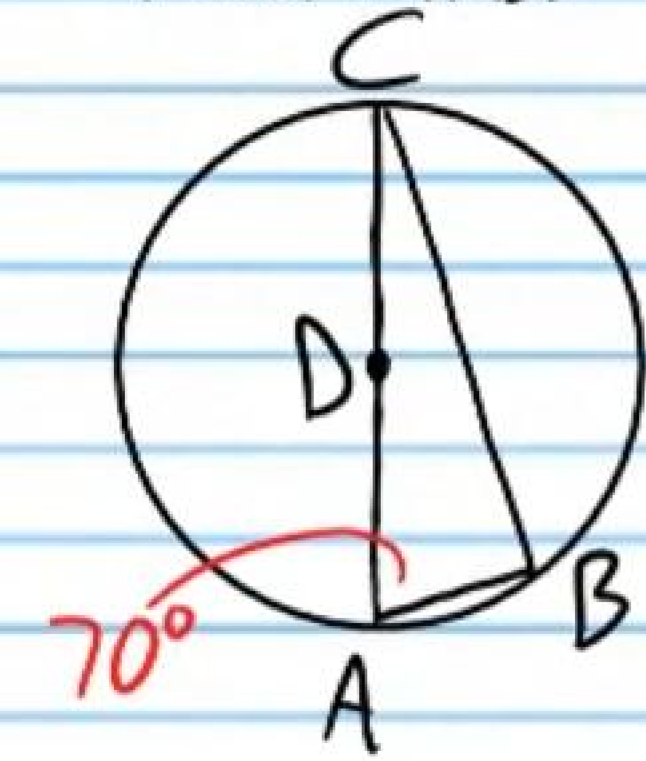
$\boxed{X = 31°}$

⑰ C is the center. Find y.

$y = 180 - 90 - 47$

$\boxed{y = 43°}$

⑱ D is the center. Find $m\widehat{AB}$.



$m\angle C = 180 - 90 - 70$

$m\angle C = 20°$

$m\widehat{AB} = 2(20)$

$\boxed{m\widehat{AB} = 40°}$

(19) E is the center. Find $m\widehat{DF}$.

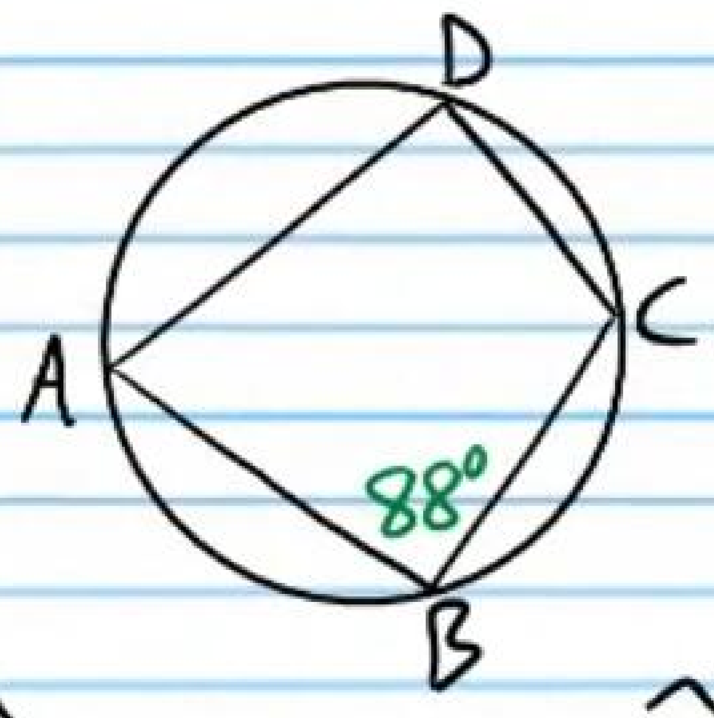
$m\angle G = 180 - 90 - 68$

$m\angle G = 22°$

$m\widehat{DF} = 2(22)$

$\boxed{m\widehat{DF} = 44°}$

(20) $m\widehat{ADC}$ & $m\angle D$



I) $88 = \frac{m\widehat{ADC}}{2}$

$\boxed{176° = m\widehat{ADC}}$

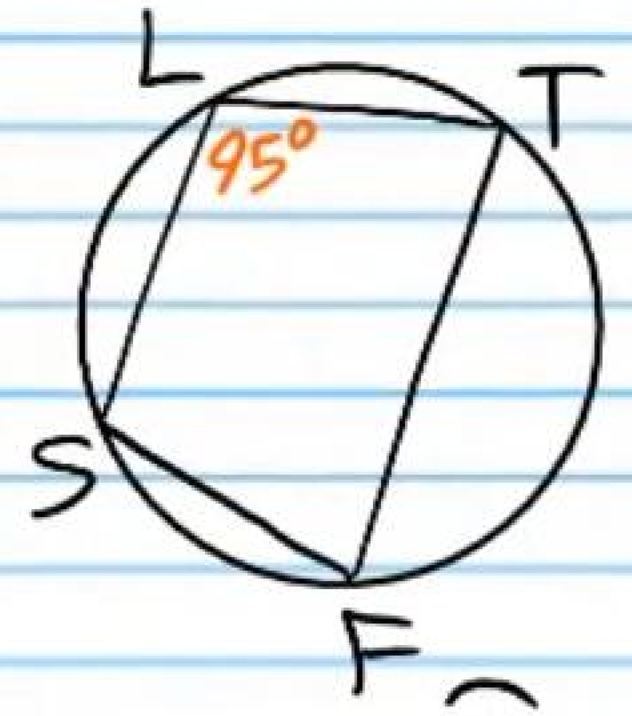
II) $m\widehat{ABC} = 360 - 176$

$m\widehat{ABC} = 184°$

$m\angle D = \frac{184}{2}$

$\boxed{m\angle D = 92°}$

(21) $m\widehat{SFT}$ & $m\angle F$.



I) $95 = \frac{m\widehat{SFT}}{2}$

$\boxed{190° = m\widehat{SFT}}$

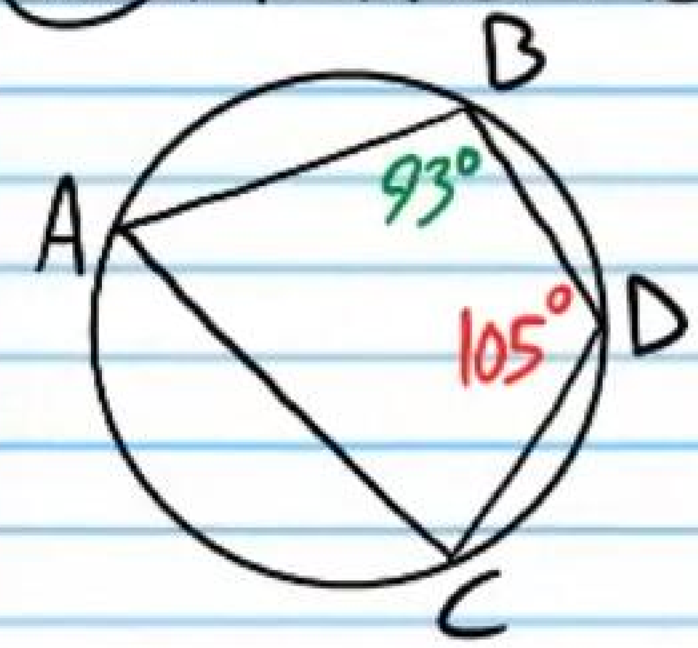
II) $m\widehat{SLT} = 360 - 190$

$m\widehat{SLT} = 170°$

$m\angle F = \frac{170}{2}$

$\boxed{m\angle F = 85°}$

(22) $m\angle A$ & $m\angle C$.



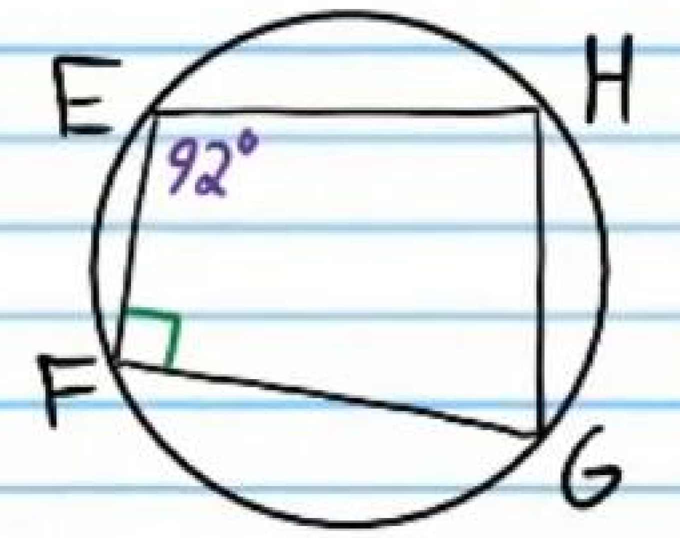
I) $m\angle A = 180 - 105$

$\boxed{m\angle A = 75°}$

II) $m\angle C = 180 - 93$

$\boxed{m\angle C = 87°}$

(23) $m\angle H$ & $m\angle G$.



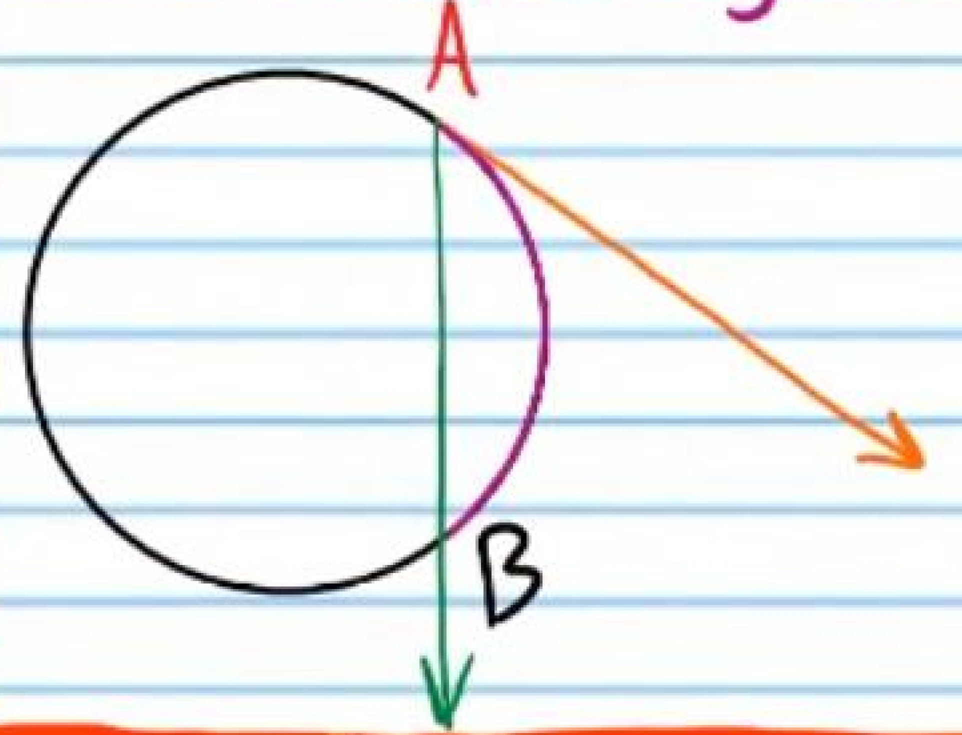
I) $m\angle H = 180 - 90$

$\boxed{m\angle H = 90°}$

II) $m\angle G = 180 - 92$

$\boxed{m\angle G = 88°}$

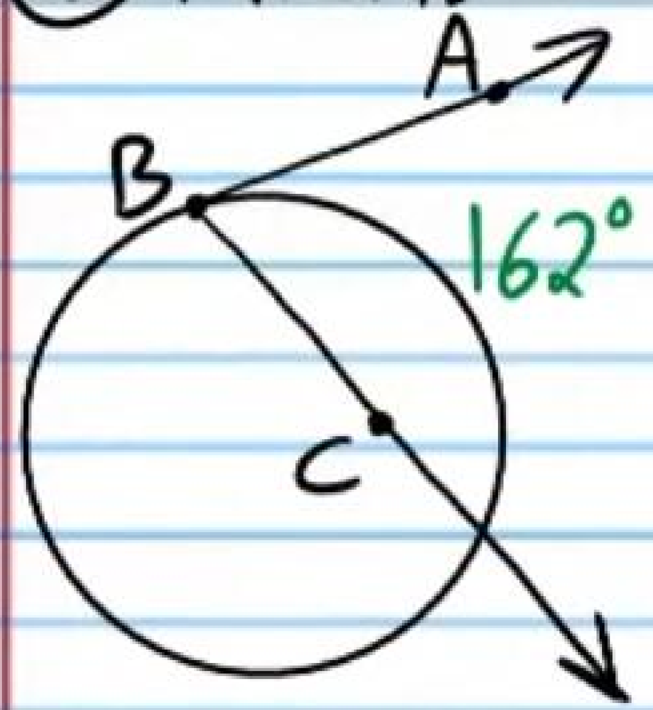
Secant-Tangent Angle Theorem: If an angle is formed by placing the vertex on a circle, and one side is extended as a secant of the circle, and the other side is extended as a tangent, then the measure of the angle is half the measure of the corresponding arc (also called the intercepted arc).



$$m\angle A = \frac{m\widehat{AB}}{2}$$

If one side of each angle below is a tangent, find the following values.

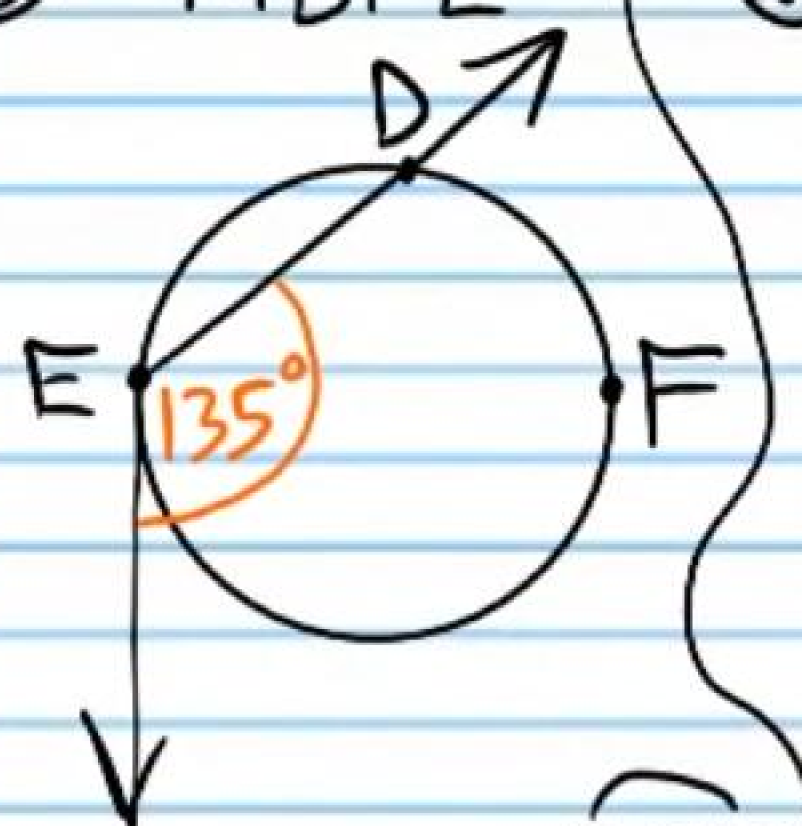
(24) $m\angle ABC$



$m\angle ABC = \frac{162}{2}$

$\boxed{m\angle ABC = 81°}$

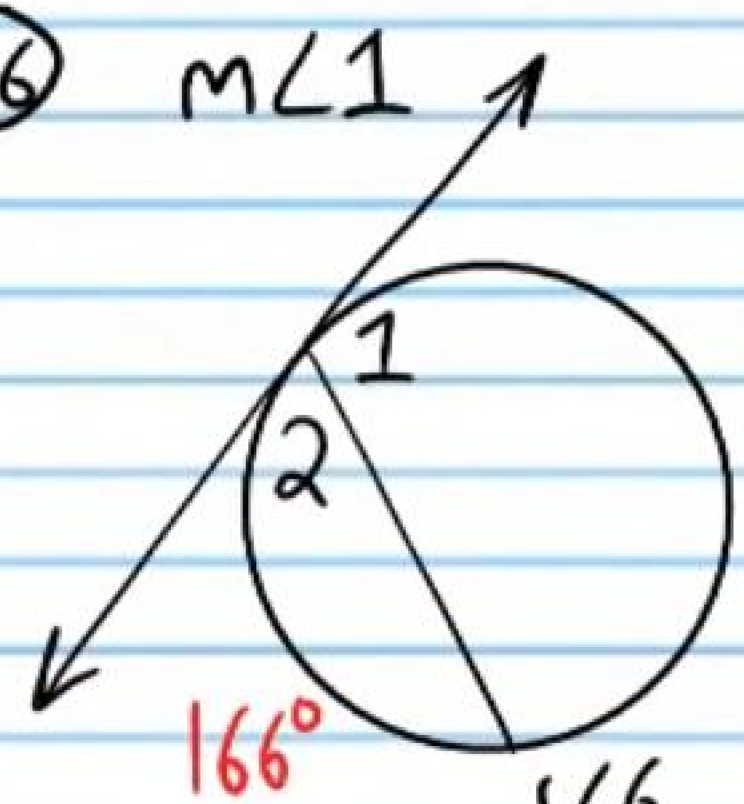
(25) $m\widehat{DFE}$



$135 = \frac{m\widehat{DFE}}{2}$

$\boxed{270° = m\widehat{DFE}}$

(26) $m\angle 1$



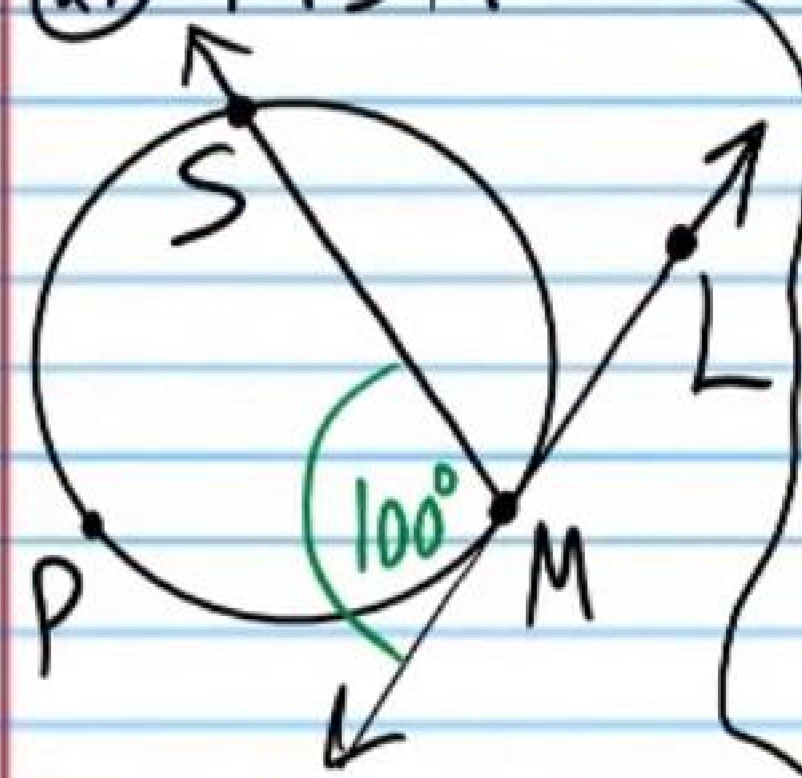
$m\angle 2 = \frac{166}{2}$

$m\angle 2 = 83°$

$m\angle 1 = 180 - 83$

$\boxed{m\angle 1 = 97°}$

(27) $m\widehat{SM}$



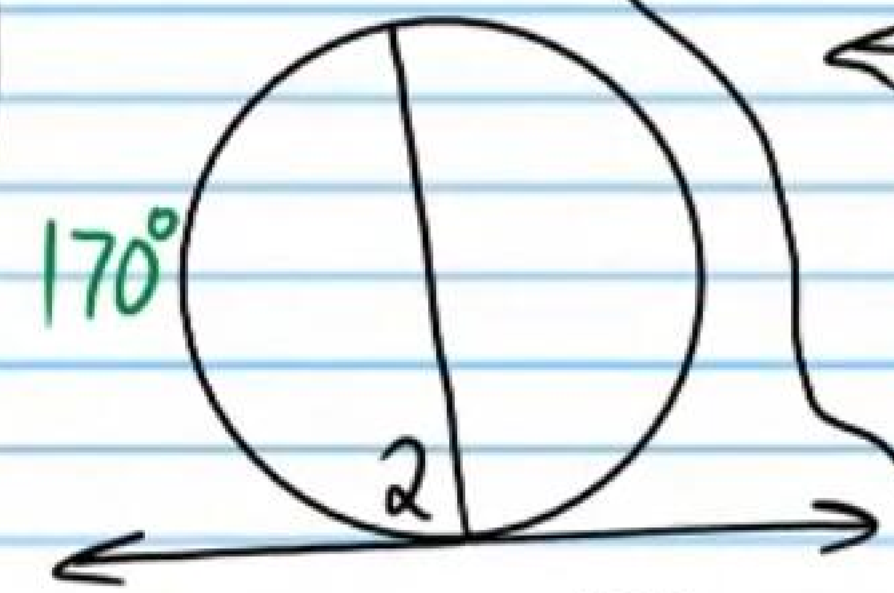
$m\angle SML = 180 - 100$

$m\angle SML = 80°$

$m\widehat{SM} = 2(80)$

$\boxed{m\widehat{SM} = 160°}$

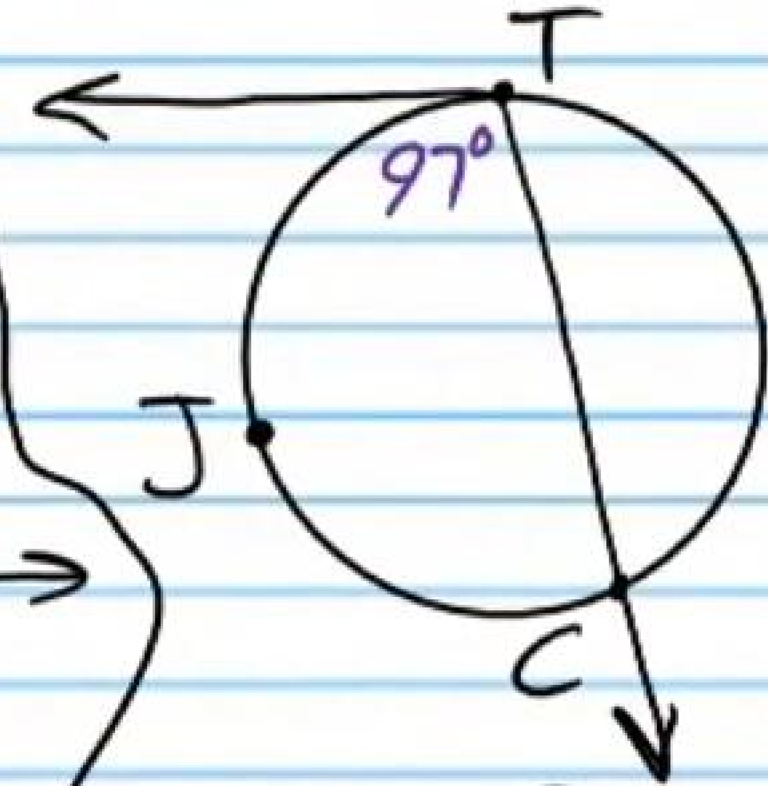
(28) $m\angle 2$



$m\angle 2 = \frac{170}{2}$

$\boxed{m\angle 2 = 85°}$

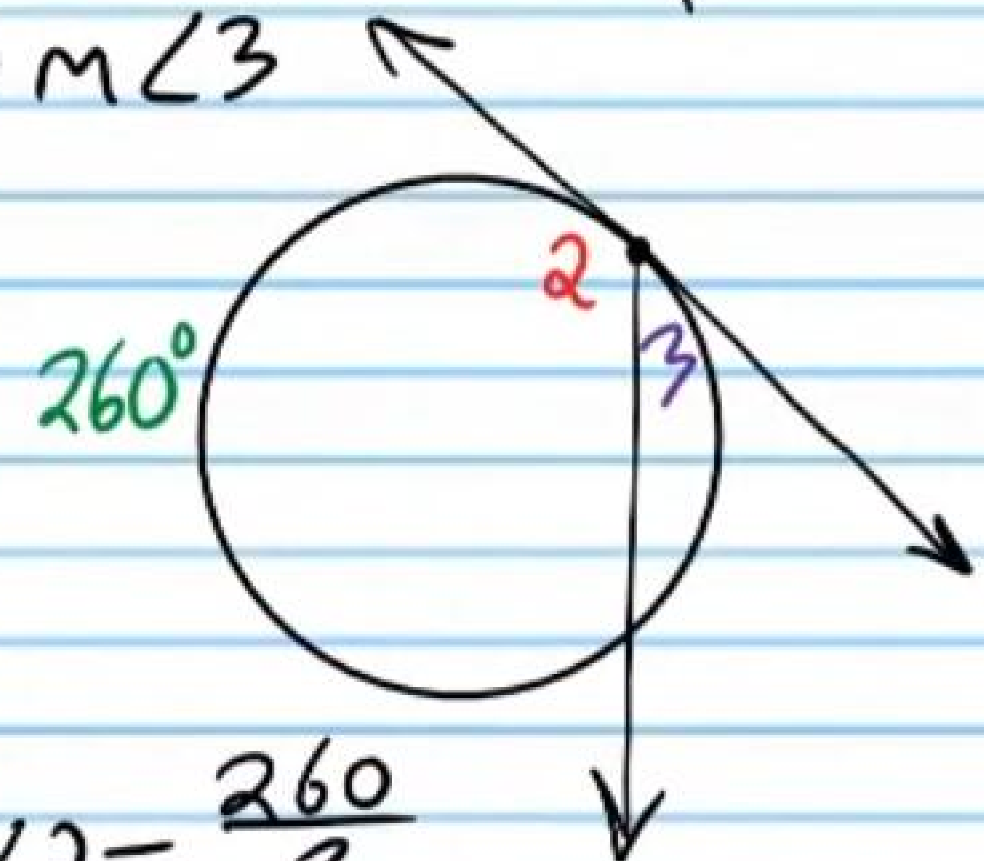
(29) $m\widehat{TJC}$



$97 = \frac{m\widehat{TJC}}{2}$

$\boxed{194° = m\widehat{TJC}}$

(30) $m\angle 3$



$m\angle 2 = \frac{260}{2}$

$m\angle 2 = 130°$

$m\angle 3 = 180 - 130$

$\boxed{m\angle 3 = 50°}$

(31) $m\widehat{MLN}$

$55 = \frac{m\widehat{MN}}{2}$

$110° = m\widehat{MN}$

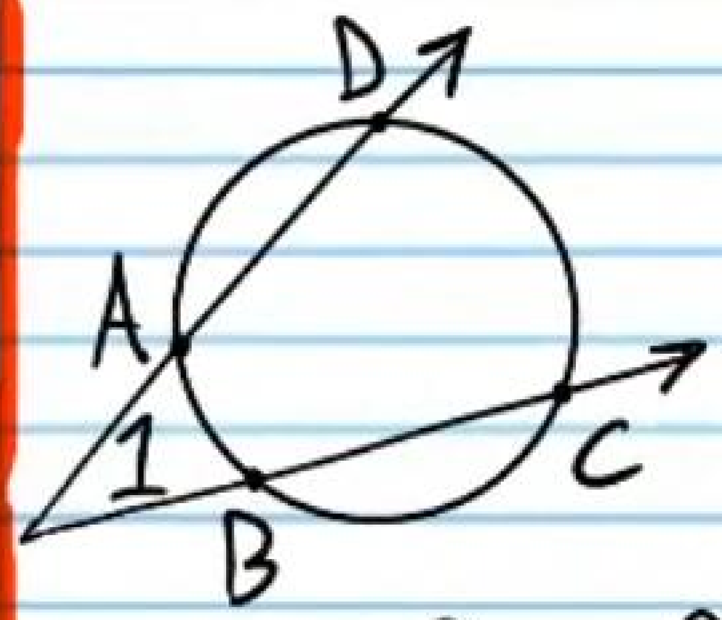
$m\widehat{MLN} = 360 - 110$

$\boxed{m\widehat{MLN} = 250°}$

Angles with Vertices Outside Circles

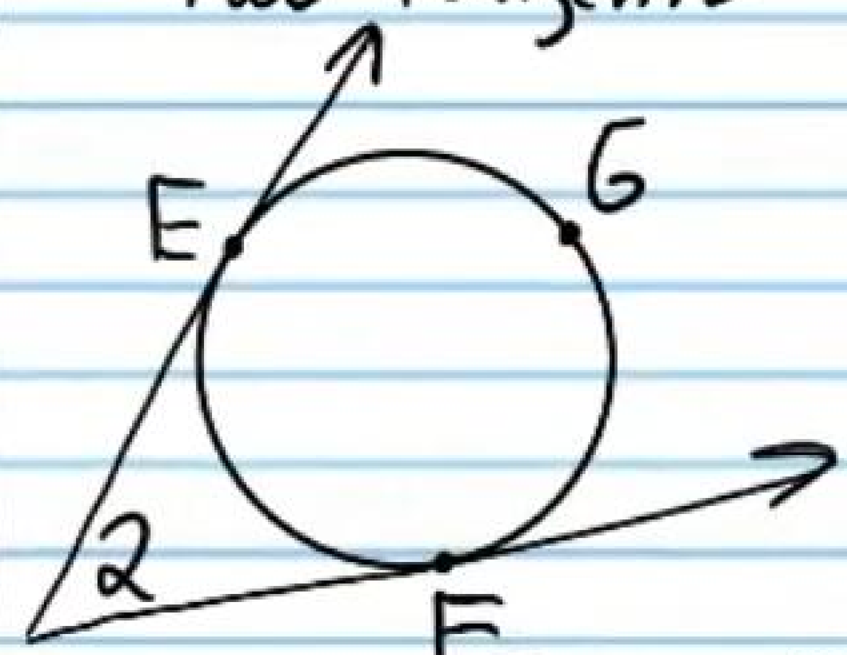
Angle Vertices Outside Circles Theorem: If an angle is formed by placing the vertex outside a circle and extending the sides so that they both intercept the circle, then the angle measure is the difference between the measures of the intercepted arcs divided by two.

Two Secants



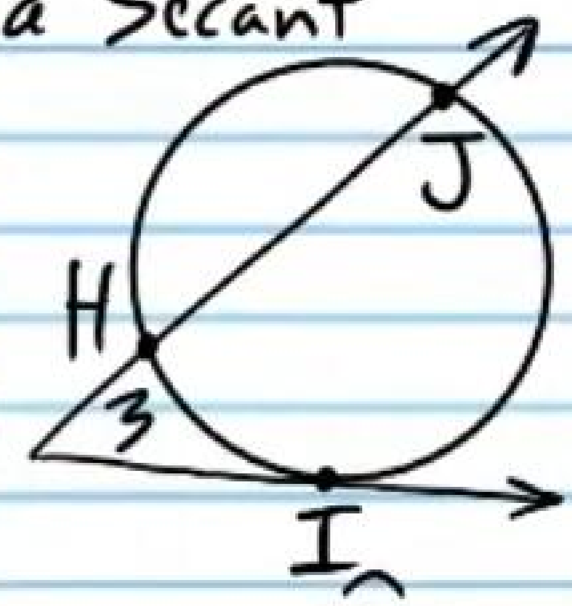
$m\angle 1 = \dfrac{m\widehat{DC} - m\widehat{AB}}{2}$

Two Tangents



$m\angle 2 = \dfrac{m\widehat{EGF} - m\widehat{EF}}{2}$

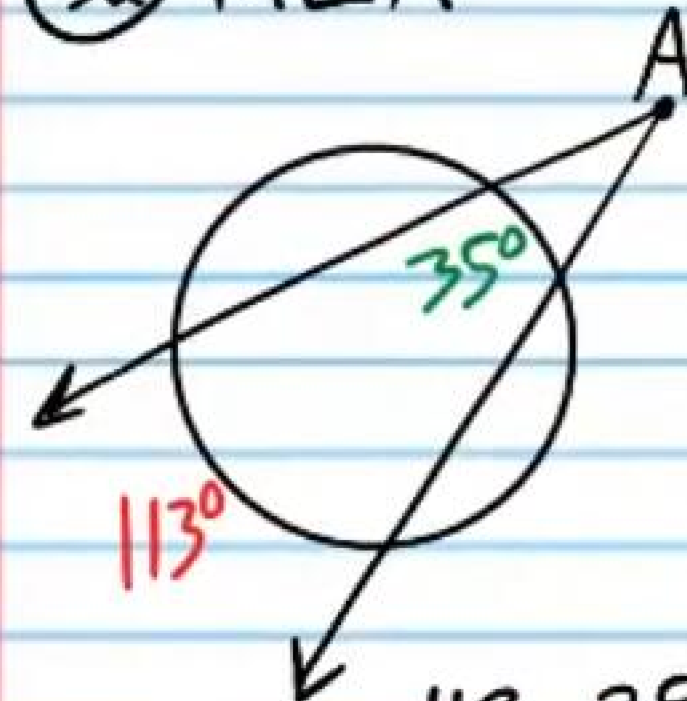
A Tangent and a Secant



$m\angle 3 = \dfrac{m\widehat{JI} - m\widehat{HI}}{2}$

Find the values below.

(32) $m\angle A$

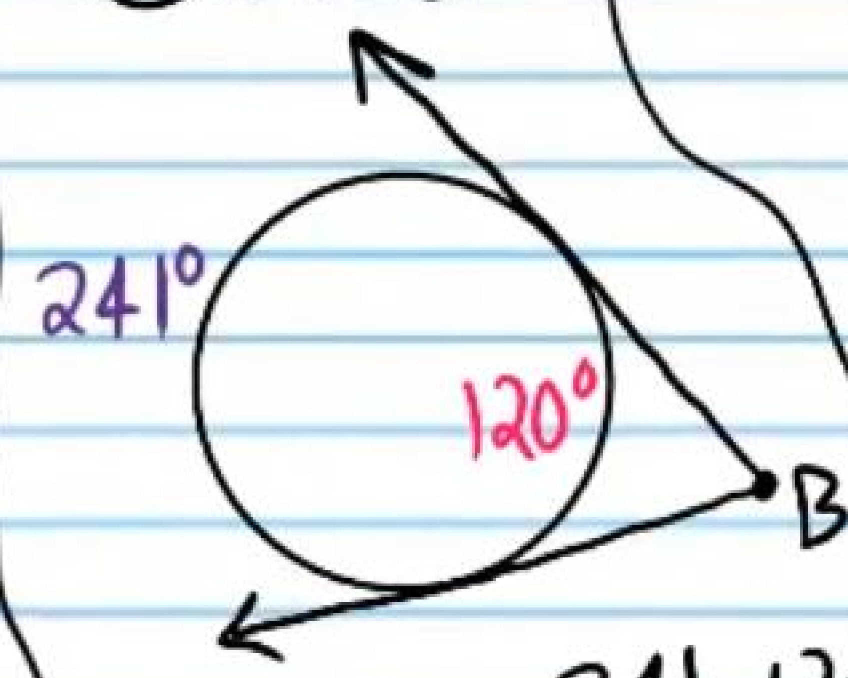


$m\angle A = \dfrac{113-35}{2}$

$m\angle A = \dfrac{78}{2}$

$\boxed{m\angle A = 39°}$

(33) $m\angle B$

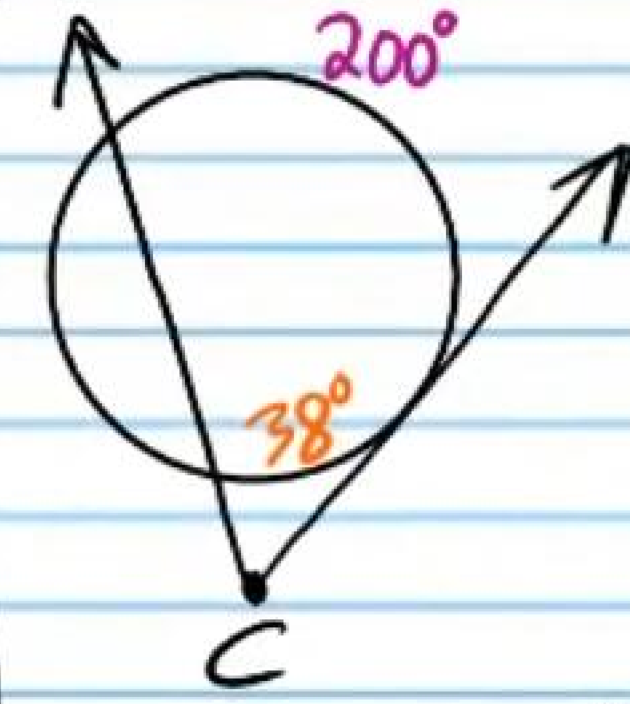


$m\angle B = \dfrac{241-120}{2}$

$m\angle B = \dfrac{121}{2}$

$\boxed{m\angle B = 60.5°}$

(34) $m\angle C$

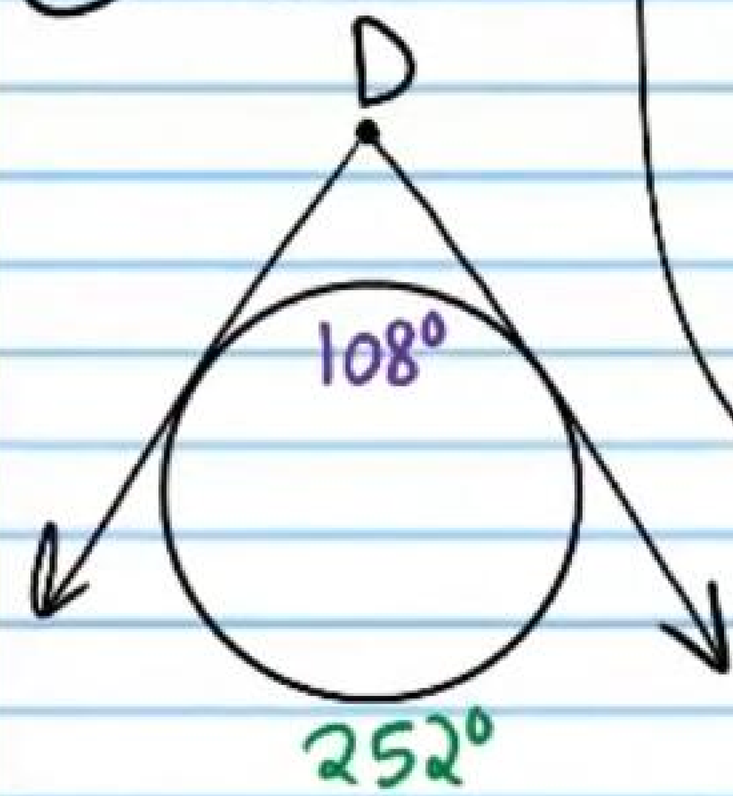


$m\angle C = \dfrac{200-38}{2}$

$m\angle C = \dfrac{162}{2}$

$\boxed{m\angle C = 81°}$

(35) $m\angle D$

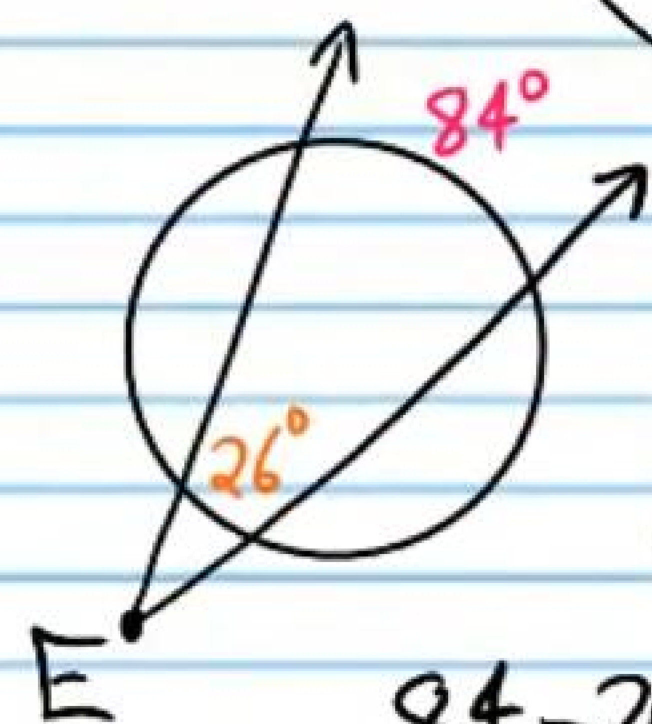


$m\angle D = \dfrac{252-108}{2}$

$m\angle D = \dfrac{144}{2}$

$\boxed{m\angle D = 72°}$

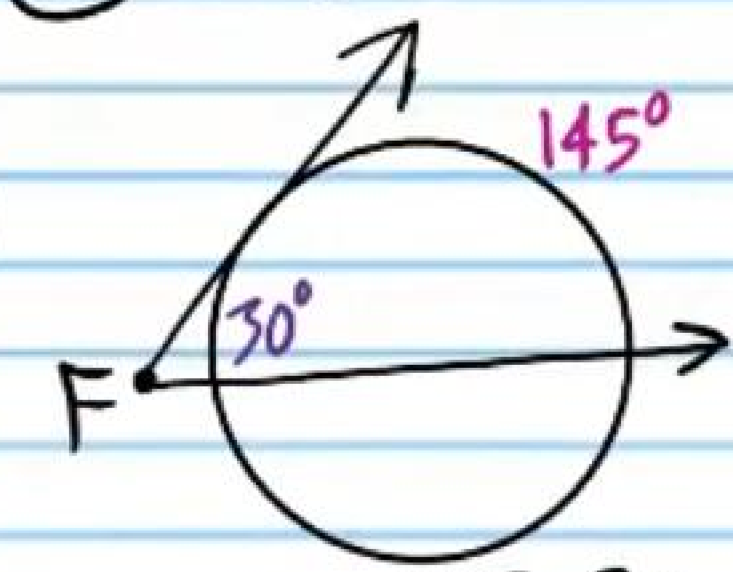
(36) $m\angle E$



$m\angle E = \dfrac{84-26}{2}$

$\boxed{m\angle E = 29°}$

(37) $m\angle F$

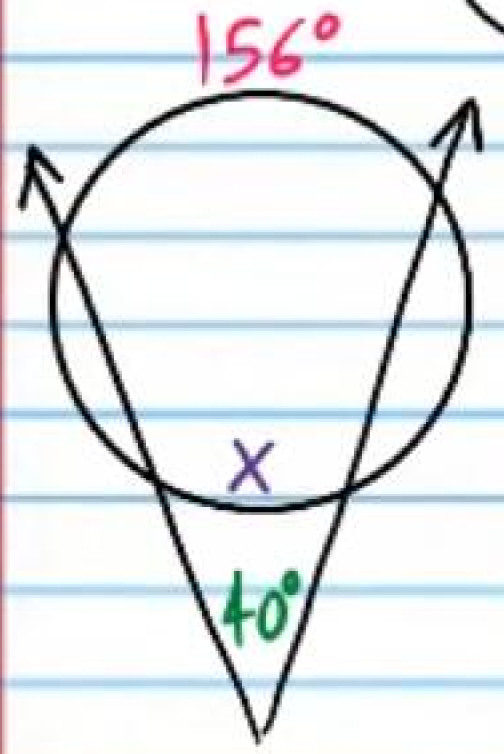


$m\angle F = \dfrac{145-30}{2}$

$m\angle F = \dfrac{115}{2}$

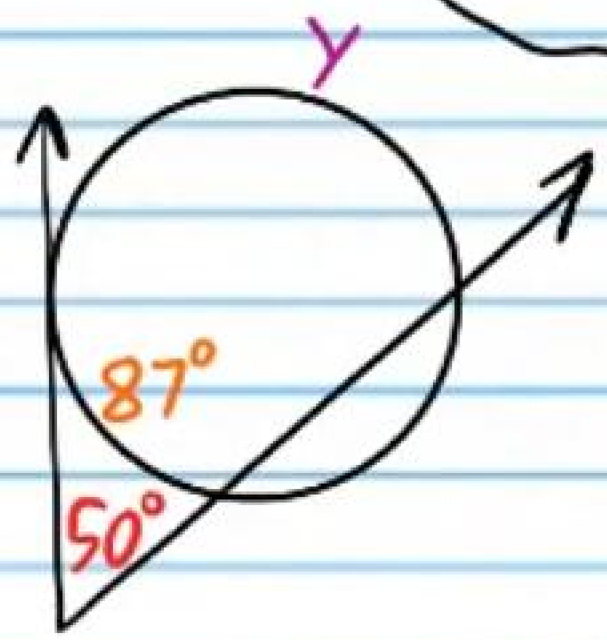
$\boxed{m\angle F = 57.5°}$

(38) x



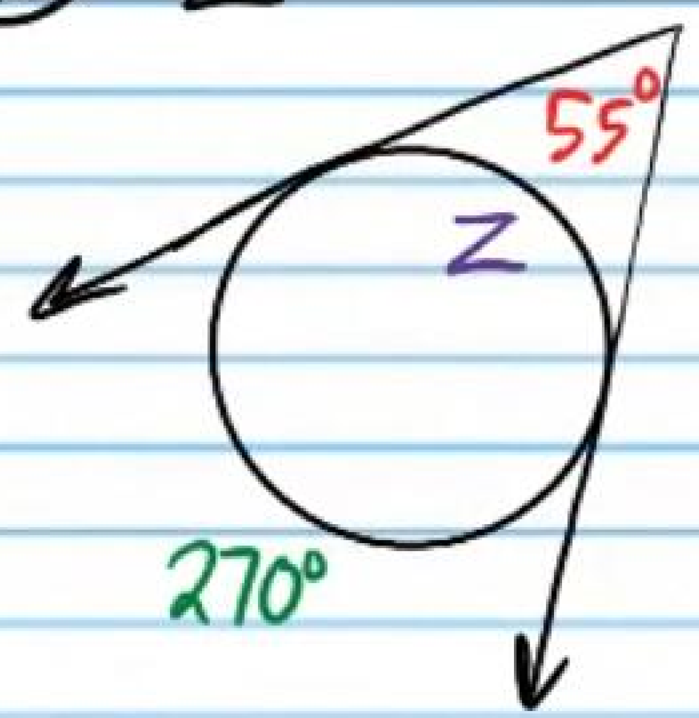
$$40 = \frac{156 - x}{2}$$
$$80 = 156 - x$$
$$-76 = -x$$
$$\boxed{x = 76°}$$

(39) y



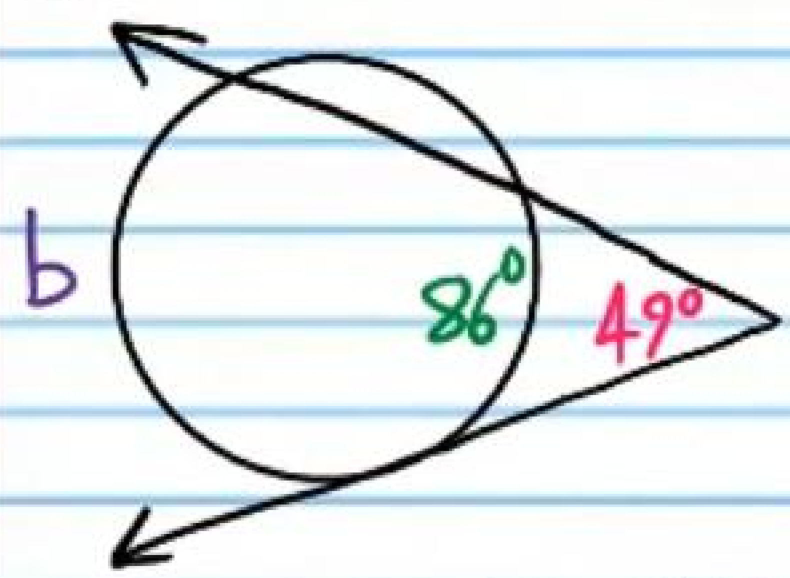
$$50 = \frac{y - 87}{2}$$
$$100 = y - 87$$
$$\boxed{187° = y}$$

(40) Z



$$55 = \frac{270 - Z}{2}$$
$$110 = 270 - Z$$
$$-160 = -Z$$
$$\boxed{160° = Z}$$

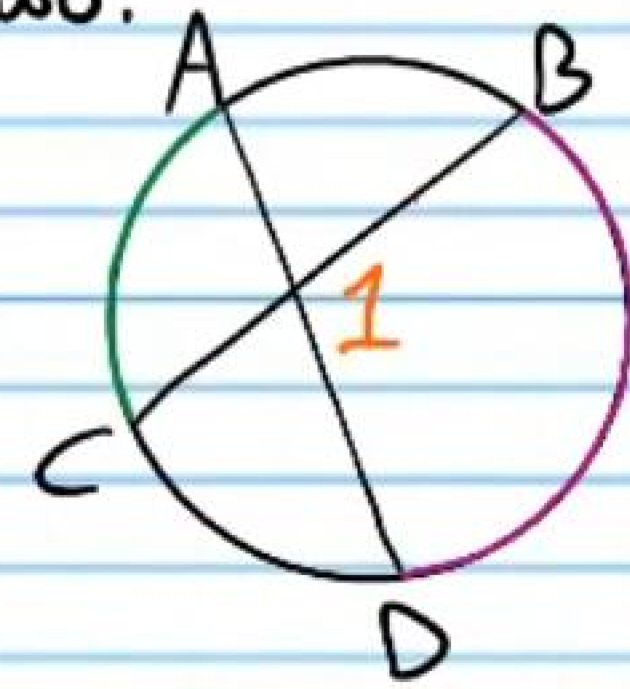
(41)



$$49 = \frac{b - 86}{2}$$
$$98 = b - 86$$
$$\boxed{184° = b}$$

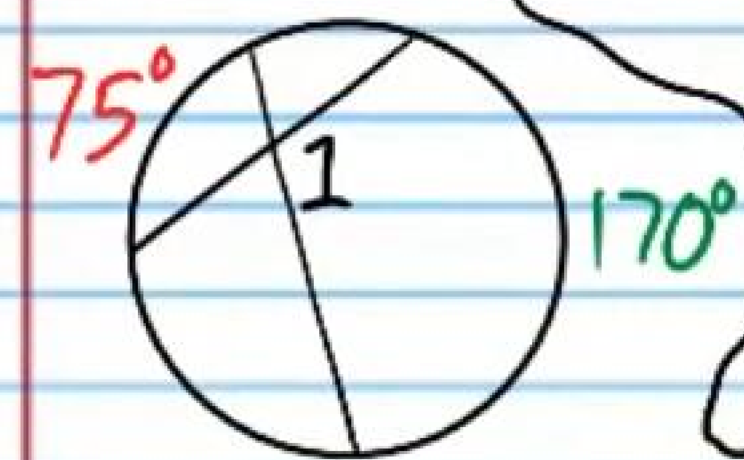
Angles with Vertices Inside Circles

Angle Vertices Inside Circles Theorem: If an angle is formed by drawing two intersecting chords inside a circle, then the measure of the angle is the sum of the measures of the intercepted arcs divided by two.



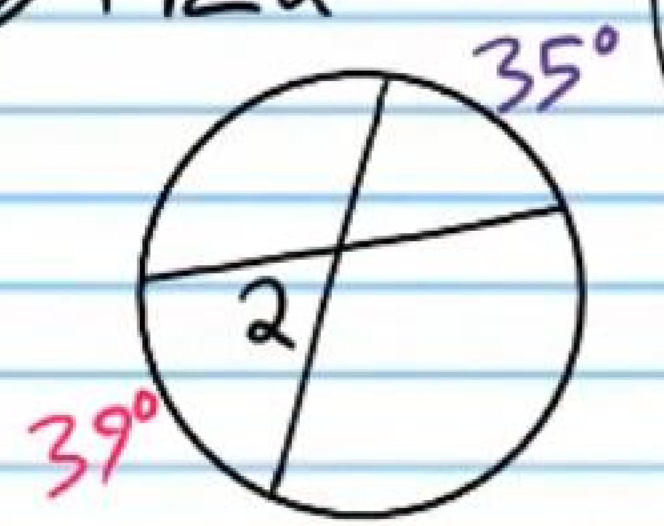
$$m\angle 1 = \frac{m\widehat{AC} + m\widehat{BD}}{2}$$

(42) $m\angle 1$



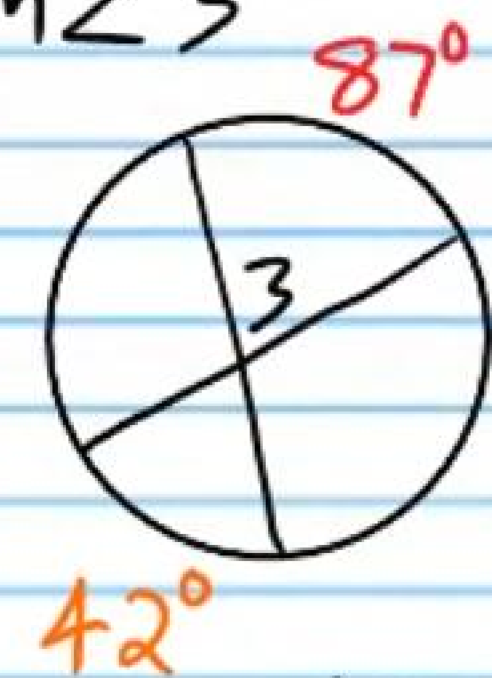
$$m\angle 1 = \frac{170 + 75}{2}$$
$$m\angle 1 = \frac{245}{2}$$
$$\boxed{m\angle 1 = 122.5°}$$

(43) $m\angle 2$



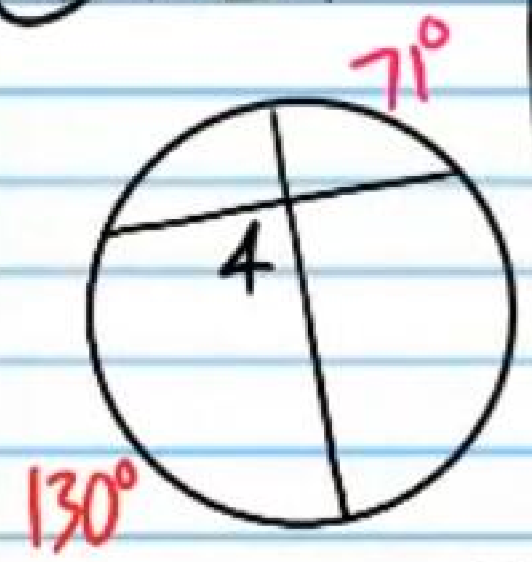
$$m\angle 2 = \frac{35 + 39}{2}$$
$$m\angle 2 = \frac{74}{2}$$
$$\boxed{m\angle 2 = 37°}$$

(44) $m\angle 3$



$$m\angle 3 = \frac{42 + 87}{2}$$
$$m\angle 3 = \frac{129}{2}$$
$$\boxed{m\angle 3 = 64.5°}$$

(45) $m\angle 4$

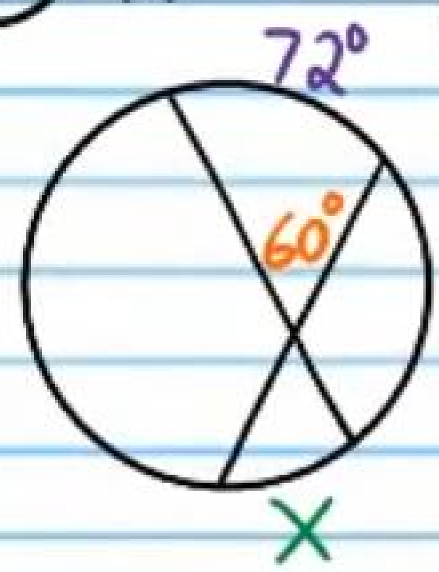


$$m\angle 4 = \frac{130+71}{2}$$

$$m\angle 4 = \frac{201}{2}$$

$$\boxed{m\angle 4 = 100.5^\circ}$$

(46) x

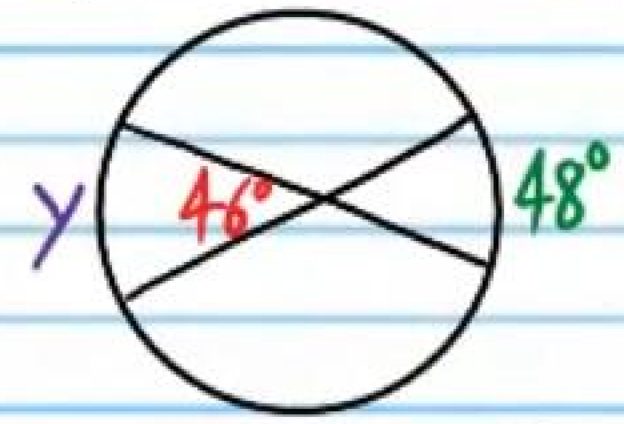


$$60 = \frac{72+x}{2}$$

$$120 = 72 + x$$

$$\boxed{48^\circ = x}$$

(47) y



$$46 = \frac{y+48}{2}$$

$$92 = y + 48$$

$$\boxed{y = 44^\circ}$$

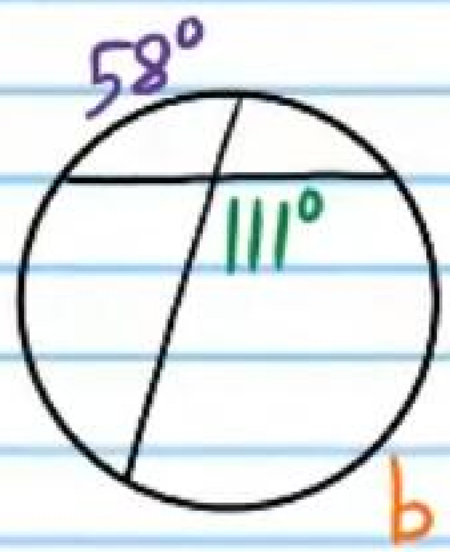
(48) a

$$80 = \frac{54+a}{2}$$

$$160 = 54 + a$$

$$\boxed{106^\circ = a}$$

(49) b



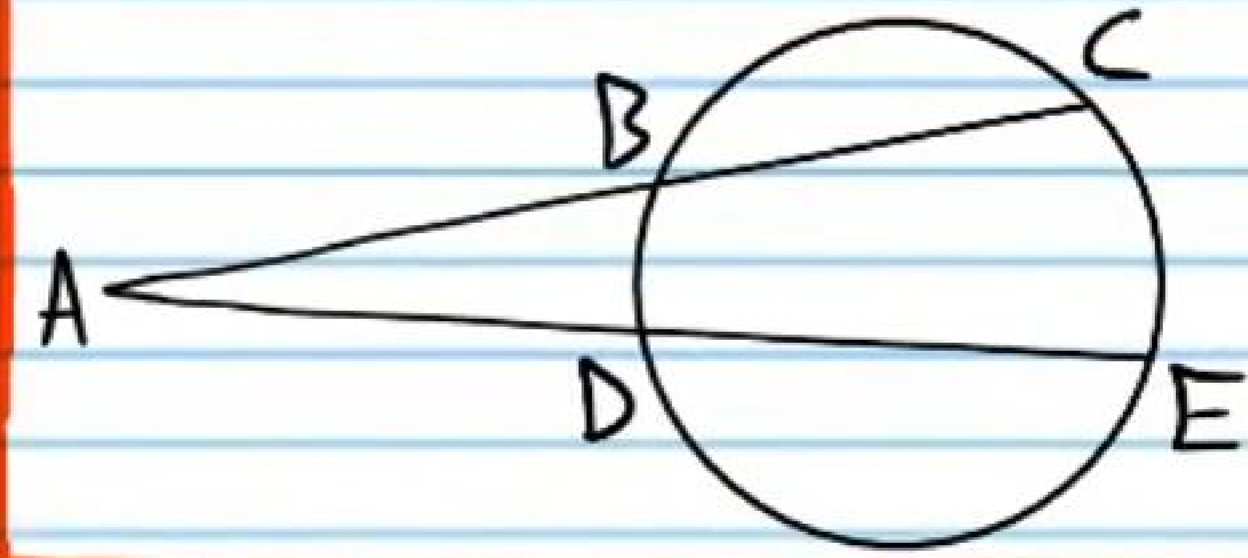
$$111 = \frac{58+b}{2}$$

$$222 = 58 + b$$

$$\boxed{164^\circ = b}$$

Segment Relationships

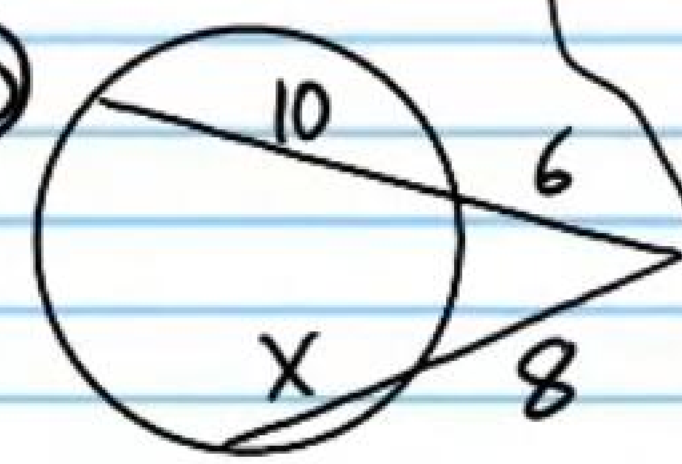
Secant Segments Theorem: If two secant segments extend from a common point outside a circle to the opposite side of the circle, then the product of the length of one segment and the length of the external part of the same segment is equal to the same type of product of the other segment.



$$AC \cdot AB = AE \cdot AD$$

Find X.

(50)



$$(10+6)6 = (8+x)8$$

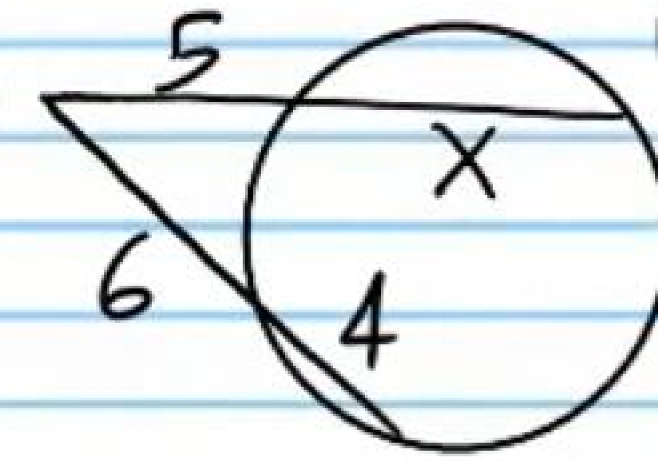
$$(16)6 = 64 + 8x$$

$$96 = 64 + 8x$$

$$32 = 8x$$

$$\boxed{4 = x}$$

(51)



$$(5+x)5 = (6+4)6$$

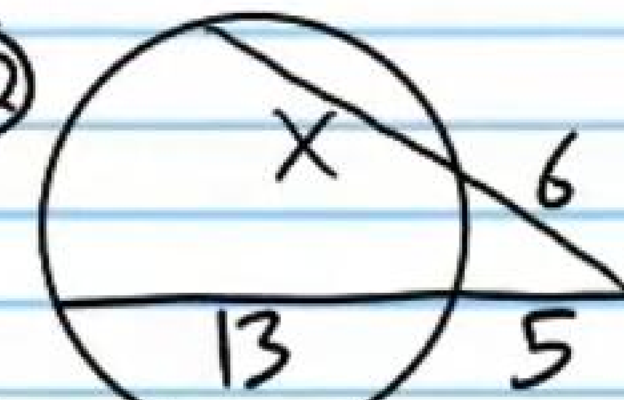
$$25 + 5x = (10)6$$

$$25 + 5x = 60$$

$$5x = 35$$

$$\boxed{x = 7}$$

(52)



$$(6+x)6 = (13+5)5$$

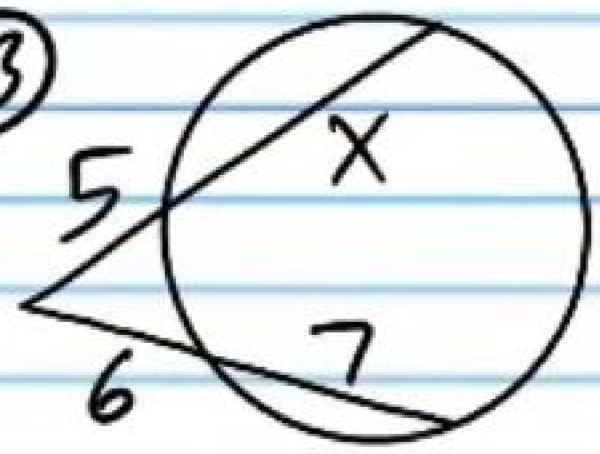
$$36 + 6x = (18)5$$

$$36 + 6x = 90$$

$$6x = 54$$

$$\boxed{x = 9}$$

53



$(5+x)5=(6+7)6$

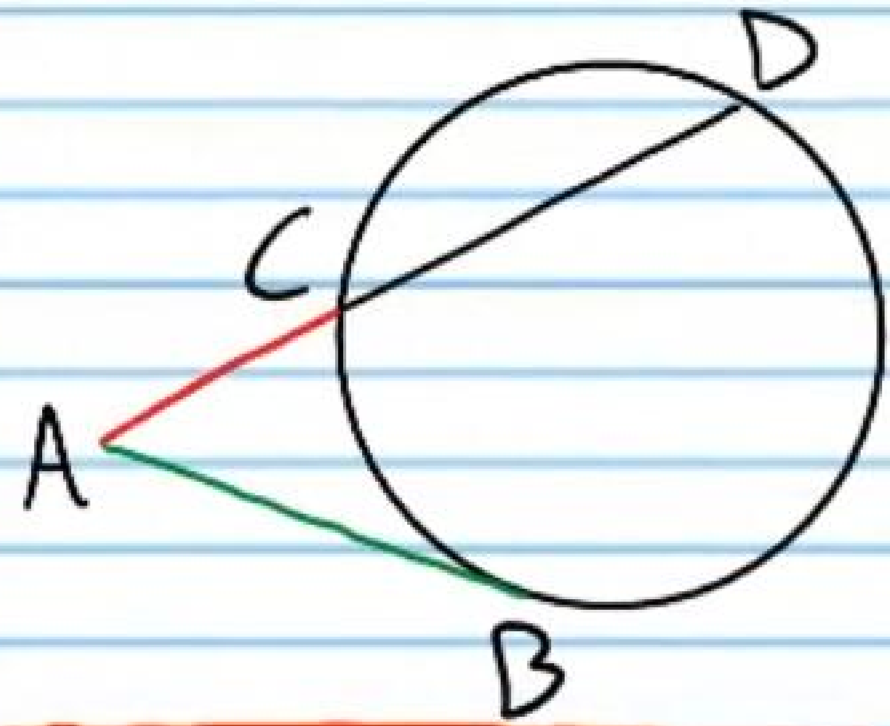
$25+5x=(13)6$

$25+5x=78$

$5x=53$

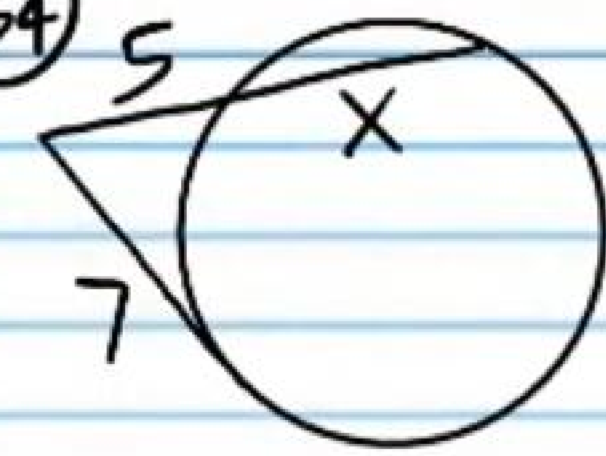
$\boxed{x=10.6}$

Secant-Tangent Segments Theorem: If a secant segment and a tangent segment extend from a common point outside a circle and terminate on the circle, then the square of the tangent length is equal to the product of the secant length and it's external length.



$(AB)^2=AC\cdot AD$

54



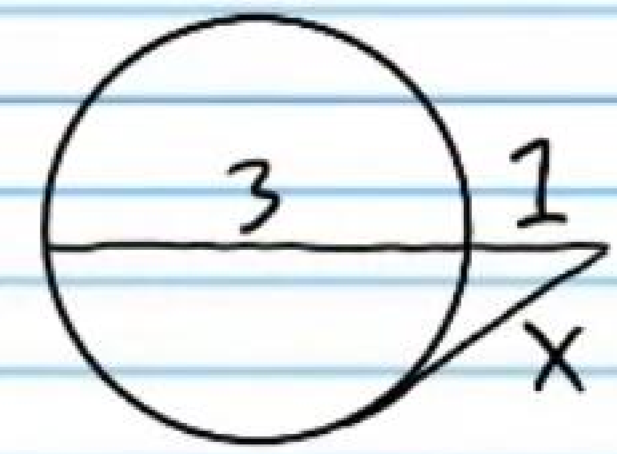
$7^2=(5+x)5$

$49=25+5x$

$24=5x$

$\boxed{4.8=x}$

55



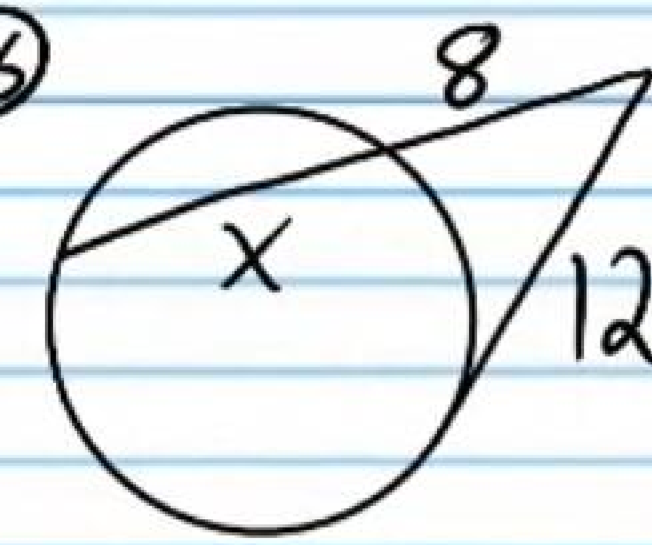
$x^2=(3+1)1$

$x^2=(4)1$

$x^2=4$

$\boxed{x=2}$

56



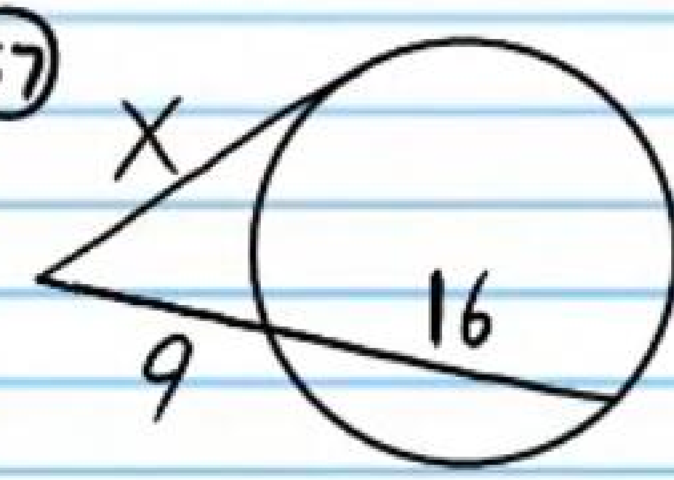
$12^2=(8+x)8$

$144=64+8x$

$80=8x$

$\boxed{10=x}$

57



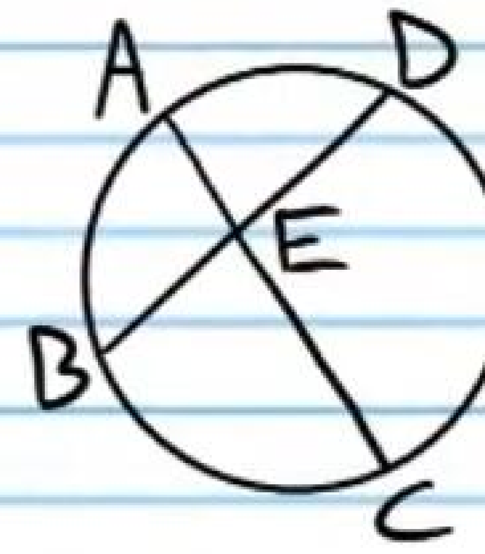
$x^2=(9+16)9$

$x^2=(25)9$

$x^2=225$

$\boxed{x=15}$

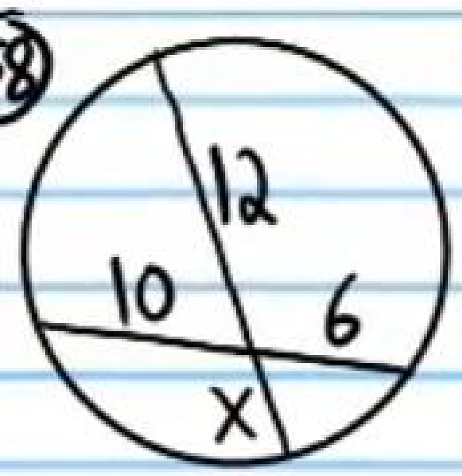
Segment Lengths of Intersecting Chords Theorem: If two chords intersect inside a circle, then the product of the resulting segments of one chord is equal to the product of the resulting segments of the other chord.



$$AE \cdot EC = BE \cdot ED$$

Find x.

58

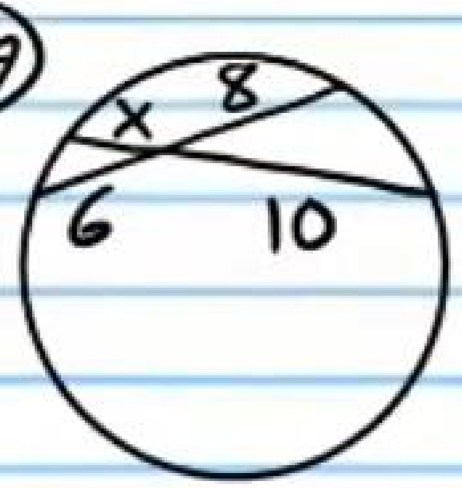


$12x = (10)6$

$12x = 60$

$\boxed{x = 5}$

59

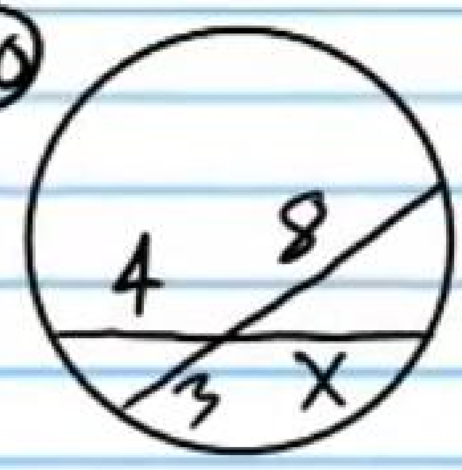


$10x = 8(6)$

$10x = 48$

$\boxed{x = 4.8}$

60

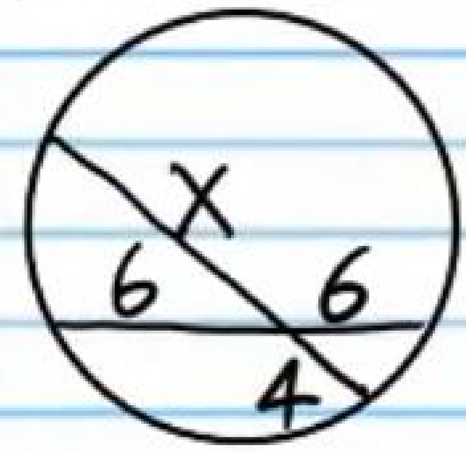


$4x = 3(8)$

$4x = 24$

$\boxed{x = 6}$

61



$4x = 6(6)$

$4x = 36$

$\boxed{x = 9}$